

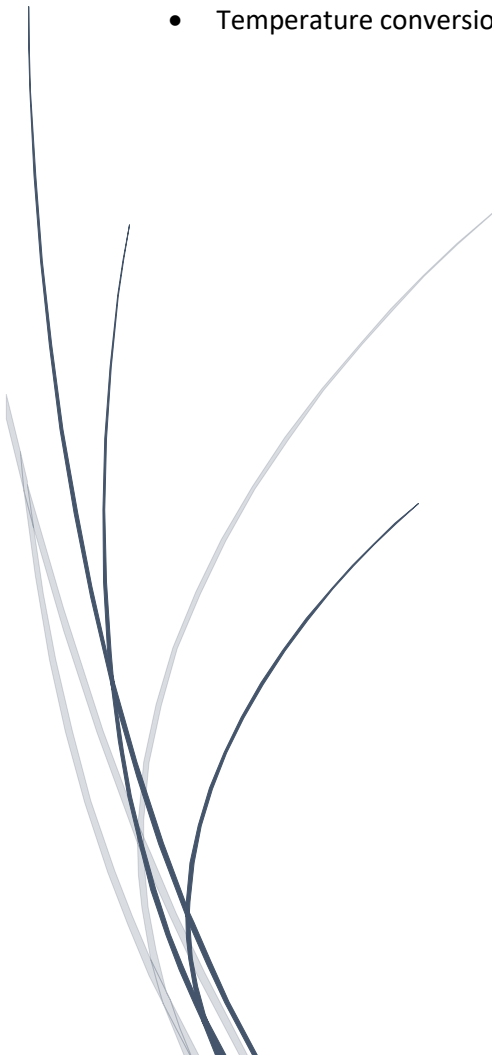
## Lab 2

### Basic syntax, In/Outputs, Data Types & Arithmetic Operators in C++

#### Learning Objectives

The following types will be covered in this lab session

- Basic syntax of C++
- Input and output using cin and cout
- Different data types
- Memory size and address of variables
- Arithmetic operations
- Temperature conversion



## 1. Basic Syntax of C++

The basic syntax of C++ defines the structure and rules of writing a C++ program. Every C++ program starts with the main() function because program execution begins from it.

A C++ program usually contains:

- Header files
- Namespace declaration
- Main function
- Statements and variables

### Syntax:

```
#include <iostream>

using namespace std;

int main() {

    // statements

    return 0;

}
```

### Example:

```
#include <iostream>    // Header file for input and output

using namespace std;  // Standard namespace

int main() {          // Main function starts

    cout << "Welcome to C++ Programming"; // Display message

    return 0;          // End of program

} return 0; }
```

## 2. Input and Output in C++

C++ uses:

- "cout" For output
- "cin" for input

Both are defined in the `<iostream>` library.

- "cout" sends data to the screen
- "cin" takes data from the user through keyboard

### Syntax:

```
cout << variable;
```

```
cin >> variable;
```

### Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int age; // Variable declaration  
    cout << "Enter your age: "; // Ask user for input  
    cin >> age; // Take input from user  
    cout << "Your age is: " << age; // Display output  
    return 0;}
```

### 3. Data Types in C++

#### if Statement:

Data types specify the type of data a variable can store. Different data types use different memory sizes.

Common data types in C++ are:

- `int` → stores integers
- `float` → stores decimal numbers
- `double` → stores large decimal numbers
- `char` → stores single character
- `string` → stores text
- `bool` → stores true or false

#### Syntax:

```
int number;      //Integer data type
float marks;     //Decimal data type
char grade;      //Character data type
```

#### Example:

```
#include <iostream>
using namespace std;
int main() {
    int age = 20;      // Integer type
    float marks = 85.5; // Float type
    char grade = 'A';  // Character type
    cout << age << endl;
    cout << marks << endl;
    cout << grade << endl;
    return 0; }
```

## 4. Arithmetic Operators in C++

Arithmetic operators are used to perform mathematical calculations in C++.

Common arithmetic operators are:

- $+$  → Addition
- $-$  → Subtraction
- $*$  → Multiplication
- $/$  → Division
- $\%$  → Modulus (remainder)

### Syntax:

```
a + b  
a - b  
a * b  
a / b  
a % b
```

### Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int a = 10, b = 3;    // Variable declaration  
    cout << "Addition: " << a + b << endl;  
    cout << "Subtraction: " << a - b << endl;  
    cout << "Multiplication: " << a * b << endl;  
    cout << "Division: " << a / b << endl;  
    cout << "Remainder: " << a % b << endl;  
    return 0;  
}
```

## 5. Output Example

Addition: 13

Subtraction: 7

Multiplication: 30

Division: 3

Remainder: 1

In this lab, we learned the basic structure of a C++ program, input/output operations, different data types, and arithmetic operators. These concepts form the foundation of C++ programming and are essential for writing basic programs and solving mathematical problems.

## Lab 2: Tasks

### Task 1: Declare Different Types of Variables & Find Their Memory Size, Location, & Values

```
#include <iostream>    // Input/output library
using namespace std;   // Standard namespace
int main() {
    int num = 10;       // Integer variable
    float marks = 85.5; // Float variable
    char grade = 'A';   // Character variable
    double pi = 3.14159; // Double variable

    // Display values
    cout << "Integer value: " << num << endl;
    cout << "Float value: " << marks << endl;
    cout << "Character value: " << grade << endl;
    cout << "Double value: " << pi << endl;

    // Display memory size using sizeof operator
    cout << "\nMemory size of int: " << sizeof(num) << " bytes\n";
    cout << "Memory size of float: " << sizeof(marks) << " bytes\n";
    cout << "Memory size of char: " << sizeof(grade) << " bytes\n";
    cout << "Memory size of double: " << sizeof(pi) << " bytes\n";

    // Display memory locations (addresses)
    cout << "\nAddress of int variable: " << &num << endl;
    cout << "Address of float variable: " << &marks << endl;
    cout << "Address of char variable: " << (void*)&grade << endl;
    cout << "Address of double variable: " << &pi << endl;

    return 0; }        // End program
```

## **Task 2: Declare and Display a Floating-Point Variable Called “pi”**

```
#include <iostream>    // Input/output library

using namespace std; // Standard namespace

int main() {

    float pi = 3.14;    // Declare floating-point variable

    cout << "Value of pi is: " << pi << endl; // Display value

    return 0;          // End program

}
```



### Task 3: Convert Temperature from Celsius to Fahrenheit

```
#include <iostream>    // Input/output library

using namespace std;    // Standard namespace

int main() {

    float celsius;        // Variable for Celsius temperature

    float fahrenheit;    // Variable for Fahrenheit temperature

    cout << "Enter temperature in Celsius: ";

    cin >> celsius;      // Take Celsius input

    // Apply conversion formula

    fahrenheit = (celsius * 9 / 5) + 32;

    // Display result

    cout << "Temperature in Fahrenheit: " << fahrenheit << endl;

    return 0;           // End program

}
```